

HIMANSHU MAITHANI

Singapore

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Professional Summary

Software Engineer with 2+ years at Fidelity and Samsung building production backend services, data pipelines, and full-stack applications. Experienced in microservices architecture (Spring Boot, FastAPI, Node.js), data engineering (Python, PySpark, PostgreSQL), and AI-integrated systems using multi-agent frameworks. Strong foundation in cloud-native infrastructure across AWS, Azure, and GKE.

Education

National University of Singapore

Master of Computing (Computer Science) — GPA: 3.85/4.0

Jan 2025 – Aug 2026

Singapore

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science Engineering — GPA: 9.48/10.0

Sep 2020 – Jun 2024

Chennai, India

Experience

SATS Ltd.

Technology Solutions Intern

May 2025 – Apr 2026

Singapore

- Built workforce compliance platform (Python, PySpark, Delta Share) cutting manual oversight by 70% for 2,000+ employees via automated data workflows, real-time alerting, and Power BI dashboards; drove on-prem-to-Azure migration using Azure Functions and Microsoft Graph API to automate cross-system data synchronization
- Built backend for a GCP-deployed internal operations platform (3,000–4,000 daily users): REST APIs for live cargo and flight data, auth, ground-handling workflow management, automated MAWB extraction from emails, and a RAG prototype for natural-language querying of operational data

Fidelity Investments

Software Development Engineer

Jan 2024 – Nov 2024

Chennai, India — Bangalore, India

- Promoted from intern to Software Development Engineer; led a 5-member team delivering a full-stack trading platform (Angular, Node.js, Spring Boot) with 85% TDD coverage, containerized as Docker/Kubernetes microservices improving deployment pipeline efficiency by 20%
- Engineered file reconciliation service processing 100K+ daily records (CSV, JSON, XML, Excel) and integrated a Python data-validation engine automating compliance report generation, cutting financial operations effort by 40%

Samsung R&D Institute India

Samsung PRISM R&D Intern

May 2023 – Nov 2023

Bangalore, India

- Designed a direct speech-to-speech translation model bypassing ASR/MT/TTS pipeline stages, achieving 40% latency reduction (2.9 PESQ, 23.5 BLEU); published findings at IEEE ICCNT 2024 and awarded Certificate of Excellence

Projects

Kiln | Python, FastAPI, AG2, PostgreSQL, Redis, Next.js, Kubernetes, Terraform

Mar 2026

- Built a self-evolving AI tool registry where a Mistral Large planning agent decomposes natural-language requests into DAGs; a synthesis agent auto-generates tool specs and hot-loads implementations in ~15 seconds across 5 FastAPI microservices with zero downtime
- Implemented MCP OAuth 2.1/PKCE authorization server enabling Claude Desktop, Cursor, and VS Code Copilot integration; deployed on GKE with Terraform IaC, Prometheus/Grafana observability, and AST-based security validation across 201 automated tests

Trade Nexus | Angular, Spring Boot, Node.js, SQLite, TypeScript

Dec 2024

- Designed a 5-tier distributed trading platform with real-time pricing engine, portfolio management, and automated advisory services; enforced security via JWT and resilience via circuit breaker patterns

Technical Skills

Languages & Frameworks: Java, Python, JavaScript, TypeScript, SQL — Spring Boot, FastAPI, Angular, React, Node.js, Next.js

Tools & Databases: Docker, Kubernetes, Terraform, AWS, Azure, PySpark, AG2, MCP, Mistral — PostgreSQL, MySQL, MongoDB, Redis

Publication: IEEE ICCNT 2024 — H. Maithani et al., *Innovations in Real-Time Speech Translation*